

Requirement Analysis

1. Introduction

The **Men's Shopping Mart – E-Commerce Database Management System** is developed to manage the daily operations of an online men's fashion store through a centralized database. The system maintains customer details, product information, inventory, suppliers, orders, payments, shipments, and customer reviews. It improves data accuracy, reduces redundancy, and supports efficient business management.

2. Business Scenario Analysis

A men's fashion retailer requires a centralized database to manage its growing online business. The system should handle customer registration, product catalog management, inventory updates, order processing, payments, shipment tracking, and customer feedback.

Currently, manual record keeping causes delays, duplicate data, and difficulty in managing business operations. The proposed database system will streamline these activities and improve overall efficiency.

3. Business Problem

The organization faces the following challenges:

- Manual maintenance of customer records.
- Duplicate and inconsistent data.
- Difficulty in managing stock levels.
- Slow order processing.
- Inefficient payment tracking.
- Limited shipment monitoring.
- Difficulty generating business reports.
- Increased chances of data entry errors.

These issues reduce productivity and affect customer satisfaction.

4. Proposed Solution

The proposed solution is the **Men's Shopping Mart – E-Commerce Database Management System**, which provides a centralized and secure database.

The system will:

- Manage customer information.
 - Organize products into categories.
 - Monitor inventory.
 - Maintain supplier records.
 - Process customer orders.
 - Record payments.
 - Track shipments.
 - Store customer reviews.
 - Generate business reports.
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5. Project Objectives

- Maintain accurate customer records.
 - Manage products and categories efficiently.
 - Store supplier information.
 - Monitor inventory.
 - Simplify order processing.
 - Record payment details.
 - Track deliveries.
 - Store customer reviews.
 - Reduce manual work.
 - Improve business efficiency.
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6. Purpose of Core Entities

Entity	Purpose
Customer	Stores customer details such as ID, name, email, phone number, address, and registration date.
Category	Stores product category information.
Product	Stores product details including price, stock quantity, and category.
Supplier	Stores supplier information.
Order	Stores customer order information.
Order Details	Stores ordered products, quantity, and unit price.
Payment	Stores payment details and payment status.
Shipment	Stores shipment information and delivery status.
Review	Stores customer ratings and product reviews.

7. Major Business Processes

1. Customer Registration

2. Category Management
 3. Product Management
 4. Inventory Management
 5. Order Processing
 6. Payment Processing
 7. Shipment Tracking
 8. Review Management
 9. Report Generation
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8. Stakeholder Analysis

Stakeholder	Responsibilities
Customer	Registers, places orders, makes payments, tracks orders, and submits reviews.
Administrator	Manages customers, products, suppliers, orders, inventory, and reports.
Supplier	Supplies products to the store.
Warehouse Staff	Updates inventory and prepares orders for dispatch.
Delivery Staff	Delivers orders and updates shipment status.
Finance Team	Records and verifies payments.
Management	Monitors reports and business performance.

9. Functional Requirements

The system shall:

- Register customers.
 - Manage customer records.
 - Manage categories and products.
 - Maintain supplier details.
 - Update inventory.
 - Process customer orders.
 - Record payment information.
 - Track shipments.
 - Store customer reviews.
 - Generate business reports.
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10. Non-Functional Requirements

The system should provide:

- Security

- Reliability
 - Performance
 - Availability
 - Data Integrity
 - Scalability
 - Maintainability
 - Backup and Recovery
 - User-Friendly Interface
 - Compatibility
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11. Expected Benefits

- Centralized data management.
 - Reduced data redundancy.
 - Better inventory control.
 - Faster order processing.
 - Accurate payment records.
 - Improved shipment tracking.
 - Better customer experience.
 - Effective business reporting.
 - Increased operational efficiency.
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12. Conclusion

The **Men's Shopping Mart – E-Commerce Database Management System** provides a centralized solution for managing online men's fashion store operations. It integrates customer management, product catalog, inventory, suppliers, orders, payments, shipments, and reviews into a single database. This requirement analysis forms the basis for database design, ER modeling, normalization, SQL implementation, and future system enhancements while ensuring efficient and reliable business operations.